

Lecture 12

Dimension argument & families w/ restricted intersection.

Given $\mathcal{F} \subseteq 2^{[n]}$,
EKR: forbid $|F \cap G| = 0$
 $\Rightarrow \leq \binom{n-1}{k-1}$, $k \geq \frac{n}{2}$
 $\rightsquigarrow$ close to 2^n

Dimension arg.

Goal • bound $|\mathcal{F}|$ from above

$\mathcal{F} \rightarrow V$ vector sp. injective
 $\text{Im}(\mathcal{F})$ linearly indep.

$|\mathcal{F}| = |\text{Im}(\mathcal{F})| \leq \dim V$

1. odd/even town

Setup. Town of n ppl, $[n]$

clubs $C_1, \dots, C_t$

$\forall i \neq j, |C_i \cap C_j| \equiv 0 \pmod{2}$

Q: if $|C_i| = \text{even}$, how many clubs we can have?

$\{1, 2\}, \{3, 4\}, \{5, 6\}, \{2^{\frac{n}{2}} - 1, 2^{\frac{n}{2}}\} \Rightarrow \geq 2^{\frac{n}{2}}$ choices

Q: What if each $|C_i| = \text{odd}$?

① $\{1\}, \{2\}, \dots, \{n\}$

② n even, $[n] \setminus \{1\}, [n] \setminus \{2\}, \dots, [n-1]$

Only linear size!

Thm Let $\mathcal{F} \subseteq 2^{[n]}$ s.t.

• $\forall F \in \mathcal{F}, |F| \equiv 1 \pmod{2}$

• $\forall F \neq F' \in \mathcal{F}, |F \cap F'| \equiv 0 \pmod{2}$

$\Rightarrow |\mathcal{F}| \leq n$

$\{0,1\}$ -vect in $\mathbb{F}_2^n$

Pf: • $\forall F \in \mathcal{F} \longrightarrow \mathbb{1}_F = (0, 1, 0, 1, \dots)$
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $1 \in F \quad 2 \in F \quad 3 \notin F \quad 4 \in F$

Claim $\mathbb{1}_F, F \in \mathcal{F}$ form orthonormal basis in $\mathbb{F}_2^n$

In particular they are linearly indep.

Pf: $\langle \mathbb{1}_F, \mathbb{1}_{F'} \rangle = \sum_{i \in F \cap F'} 1 = |F \cap F'| = \begin{cases} 0 & F \neq F' \\ 1 & F = F' \end{cases}$

$|\mathcal{F}| = |\{\mathbb{1}_F, F \in \mathcal{F}\}| \leq \dim \mathbb{F}_2^n = n$



2. Sets w/ ≤ 2 distinct distances

• Given $X \subseteq \mathbb{R}^n$, if all pairs of pts in X give the same distance, $|X| \leq ?$

$X \longrightarrow$ reg simplex $|X| \leq n+1$



$\mathbb{R}^2$

Q: What if two distances allowed?

Const: $X = \{e_i + e_j ; i, j \in \binom{[n]}{2}\}$
 $|X| = \binom{n}{2}$ $\uparrow$ standard basis

Open: Max. # of pts w/ two distances?

Best: $|X| \leq \binom{n+2}{2}$

Thm $X \subseteq \mathbb{R}^2$ w/ $\leq$ two distances

$$\Rightarrow |X| \leq \frac{1}{2}(n+2)(n+1)$$

Pf: $\left(\begin{array}{l} \text{write} \\ \|x\| = \|x\|_{\ell_2} = \sqrt{\sum x_i^2} \end{array} \right)$

Let $X = \{v_1, \dots, v_t\}$, with distances d, D .

$v_i \in X \longrightarrow$ n -variate $\deg-4$ $P_i(x) = (\|x - v_i\|^2 - d^2)$
 $\# \text{ monomials } n\text{-variable } \deg-4 \in O(n^4)$ $\text{poly. } \forall x \in \mathbb{R}^n \quad (\|x - v_i\|^2 - D^2)$

Claim $P_i, i \in [t]$, are linearly indep.

Pf: • Fix $i \neq j$.

$$P_i(v_j) = (\|v_j - v_i\|^2 - d^2)(\|v_j - v_i\|^2 - D^2) = 0$$

$$\bullet P_i(v_i) = d^2 D^2 \neq 0$$

$$\bullet \sum \alpha_i P_i \equiv 0$$

$$\sum \alpha_i P_i(v_j) = \alpha_j P_j(v_j) = 0 \Rightarrow \alpha_j = 0$$



Note that

$$\|x - v_i\|^2 = \sum_{j \in [n]} (x_j - v_{ij})^2$$

$$= \sum_{j \in [n]} x_j^2 - 2 \sum_j x_j v_{ij} + \sum_j v_{ij}^2$$

$$P_i(x) = (\|x - v_i\|^2 - d^2) \cdot (\|x - v_i\|^2 - D^2)$$

Spanned by

- $(\sum x_j^2)^2$ 1 choice
- $(\sum x_j^2) \cdot x_k$ n choices
- $x_j \cdot x_k \quad j \leq k$ $(n+1)/2$
- x_j n choices
- 1 constant term 1 choice

$$\dim \leq \frac{1}{2}(n+4)(n+1)$$



———— 3. sets w/ restricted intersections ————

Thm (Frankl - Wilson) [FW]:

Let $\mathcal{F} \subseteq 2^{[n]}$, p odd prime.

$$|\mathcal{F}| \leq \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{p-1}$$

• $\forall F \in \mathcal{F}, |F| \equiv 0 \pmod p \Rightarrow$

• $\forall F \neq F' \in \mathcal{F}, |F \cap F'| \not\equiv 0 \pmod p$

Rank Consider $n=4p$, $\mathcal{F} \subseteq \binom{[4p]}{2p} = \binom{[n]}{n/2}$

& $\forall F \neq F', |F \cap F'| \neq p = n/4$

$$\Rightarrow |I| \leq \sum_{i=0}^{p-1} \binom{n}{i} \leq \frac{n}{4} \cdot \binom{n}{n/4} \leq 2^{(1-\epsilon)n} \quad \text{Exp. Small!}$$

pf Associate to each $\forall x \in \mathbb{F}_p^n$

$$F \in \mathcal{F} \longrightarrow P_F(x) = 1 - \left(\sum_{i \in F} x_i \right)^{p-1}$$

n-var.
deg-(p-1)

Claim $P_F, F \in \mathcal{F}$, linearly indep.

$$\text{pf: } \left[\begin{array}{l} \text{NTS } F \neq F' \in \mathcal{F} \quad P_F(\mathbb{1}_{F'}) = 0 \\ P_F(\mathbb{1}_F) = 1 \quad (\text{over } \mathbb{F}_p) \end{array} \right]$$

$$P_F(\mathbb{1}_{F'}) = 1 - |F \cap F'|^{p-1} \stackrel{\text{FLT}}{=} \begin{cases} 0 & \text{if } F \neq F' \\ 1 & \text{if } F = F' \end{cases}$$

$|F \cap F'| \not\equiv 0 \pmod p$
 $|F \cap F'| \equiv 0 \pmod p$

$$\sum \alpha_F P_F \equiv 0, \quad \sum \alpha_F P_F(\mathbb{1}_{F'}) = \alpha_{F'} \equiv 0$$

Multi-linearisation (Further reduce dim)

$$x^r \longrightarrow x \quad \text{same over } \{0,1\}$$

$$x_1^3 x_2^5 x_3^7 \longrightarrow x_1 x_2 x_3$$

$$P_F \longrightarrow \tilde{P}_F \quad \text{multilinear} \quad \leftarrow \begin{array}{l} \text{also} \\ \text{linearly} \\ \text{indep.} \end{array}$$

$$\text{Again } \tilde{P}_F(\mathbb{1}_{F'}) = \begin{cases} 1 & F = F' \\ 0 & F \neq F' \end{cases}$$

• spanned by n-var. monomials of deg (p-1)

$$\dim \leq \# \text{ mono.} \leq \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{p-1} \quad \square$$

Applications

Con 1: p odd prime, $n=4p$. $X \in \mathcal{S}^{n-1} \subseteq \mathbb{R}^n$

if X has no orthogonal pairs
(no $\langle x, y \rangle = 0$) $\Rightarrow \mu(X)$ exp. small

Idea: • Discretise the problem by considering a (disc) set of unit vectors

orthog pairs $\Leftrightarrow$ pairs w./ forbidden intersection

• averaging

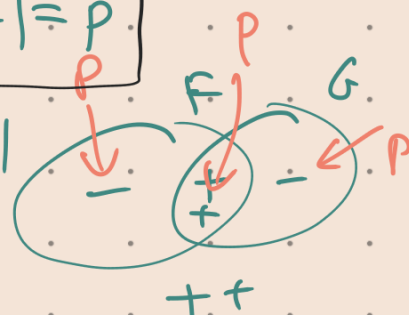
Pf: • $\mathcal{F} = \binom{[n]}{n/2} = \binom{[4p]}{2p}$ $i \in F \quad j \notin F$
 $F \in \mathcal{F} \mapsto v_F = (\dots, \frac{1}{\sqrt{n}}, \dots, -\frac{1}{\sqrt{n}}, \dots)$

$$L = \{v_F : F \in \mathcal{F}\} = \frac{1}{\sqrt{n}} (\mathbb{1}_F - \mathbb{1}_{[n] \setminus F})$$

Claim $\forall v_F, v_G \in L$,

$$\langle v_F, v_G \rangle = 0 \Leftrightarrow |F \cap G| = p$$

Pf: $\langle v_F, v_G \rangle = \frac{1}{n} (|F \cap G| - |F \Delta G| + |[n] \setminus (F \cup G)|)$



$$= \frac{1}{n} (n - 2|F \Delta G|) = 1 - \frac{2}{n} |F \Delta G|$$

Claim

$\Rightarrow$

$$|X \cap L| \leq \sum_{i=0}^{p-1} \binom{n}{i} \text{ exp small } \leq 2^{(1-\epsilon)n}$$

" $2p = n/2$ "

⊢

as $X \cap L$ corresp. to $\mathcal{F}' \subseteq \mathcal{F}$ with
no $|\mathcal{F} \cap \mathcal{F}'| = p$.

- Take a unif random rotation ρ

$$\mathbb{P}(\text{a fixed pts in } S^{n-1} \in \rho X) = \mu(X)$$

$$\Rightarrow \mathbb{E} |\rho X \cap L| = \mu(X) \cdot |L| = \mu(X) \cdot \binom{n}{n/2}$$

On the other hand $\forall \rho, | \rho X \cap L | \leq \sum_{i=0}^{p-1} \binom{n}{i}$

$$\mu(X) \cdot \binom{n}{n/2} \leq \sum_{i=0}^{p-1} \binom{n}{i} \quad \square$$

Borsuk's conj
1933

Conj $\forall$ bdd $X \subseteq \mathbb{R}^n$ can be covered by
 $\leq (n+1)$ pieces of small diameter.

(Kahn-Kalai 93) # pieces needed $\geq \exp.$ many
(smaller)

smaller
pieces needed $\geq \frac{|X|}{\sum_{i=0}^{n/2} \binom{n}{i}} \geq 2^{cn}$

Cor Let p odd prime, $n=4p$. $\exists X \subseteq \mathbb{R}^{n^2}$ of size

$\binom{n}{n/2}$ s.t. $\forall Y \subseteq X, \text{diam } Y < \text{diam } X$
 $\Rightarrow |Y| \leq \sum_{i=0}^{p-1} \binom{n}{i} \leq 2^{(1-c)n}$

Idea: • Take tensor $\Rightarrow$ distance maximised
when pair orthogonal

• So subset of small diam. contain no
Orthog. pair:

pf: $\mathcal{F} = \left(\begin{bmatrix} n \\ n/2 \end{bmatrix} \right)$, $v_F = \frac{1}{\sqrt{n}} (\dots, \overset{i \in F}{\underset{\downarrow}{1}}, \dots, \overset{j \notin F}{\underset{\downarrow}{-1}}, \dots)$

$$L = \{ v_F = \frac{1}{\sqrt{n}} (1_F - 1_{[n] \setminus F}) : F \in \mathcal{F} \}$$

$$X = \{ v \otimes v = v \cdot v^T : v \in L \} \subseteq \mathbb{R}^{n^2}$$

Claim $\text{diam } X = \sqrt{2}$
 $\| v \otimes v - w \otimes w \|^2 \leq 2$ with
equality iff $\langle v, w \rangle = 0$

Claim $\Rightarrow \forall Y \text{ diam}(Y) < \text{diam } X = \sqrt{2}$

$\Rightarrow Y$ has no $v_F \otimes v_F, v_G \otimes v_G$ s.t.
 $\langle v_F, v_G \rangle = 0 \Leftrightarrow \text{no } |F \cap G| = p$

(in previous pf)

$v \otimes v = v \cdot v^T = \begin{pmatrix} v_1 & \dots & v_n \\ \vdots & \ddots & \vdots \\ v_1 & \dots & v_n \end{pmatrix}$ $\Rightarrow |Y| \leq \sum_{i=0}^{p-1} \binom{n}{i}$

pf of claim $\langle \underline{v \otimes v}, \underline{w \otimes w} \rangle = \sum_{i,j} v_i v_j w_i w_j$

$$= \left(\sum_i v_i w_i \right) \left(\sum_j v_j w_j \right) = \langle v, w \rangle \langle v, w \rangle$$

$$= \underline{\langle v, w \rangle^2} \geq 0 \quad \text{equality iff } \langle v, w \rangle = 0$$

Also $\|v\|^2 = 1$, $\|v \otimes v\|^2 = \|v\|^4 = 1$

$$\|v \otimes v - w \otimes w\|^2 = \underbrace{\|v \otimes v\|^2}_{=2} + \underbrace{\|w \otimes w\|^2}_{=2} - 2\langle v \otimes v, w \otimes w \rangle$$

$$= 2 - 2\langle v, w \rangle^2 \leq 2$$

"=" iff $\langle v, w \rangle = 0$ 

